

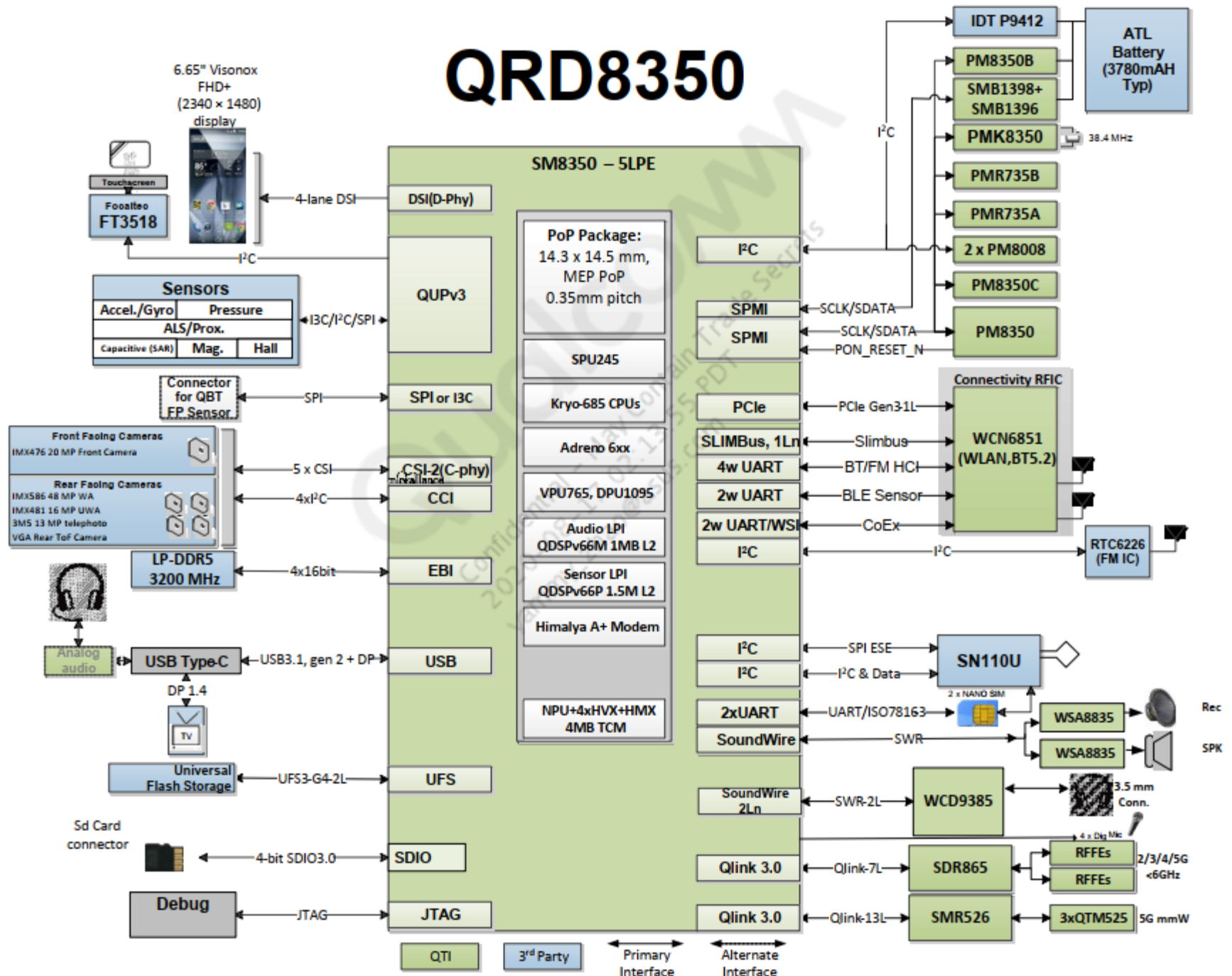
ZS672KS Vodka Baseband Trouble shooting

Agenda

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- System Block Diagram
- Phone Introduction
- PCB Introduction
 - Main Board (MB)
 - SIM Sub-Board (SIM SB)
 - USB Sub-Board (USB SB)
 - Camera Sub-Board (CAM SB)
- Repair Guideline

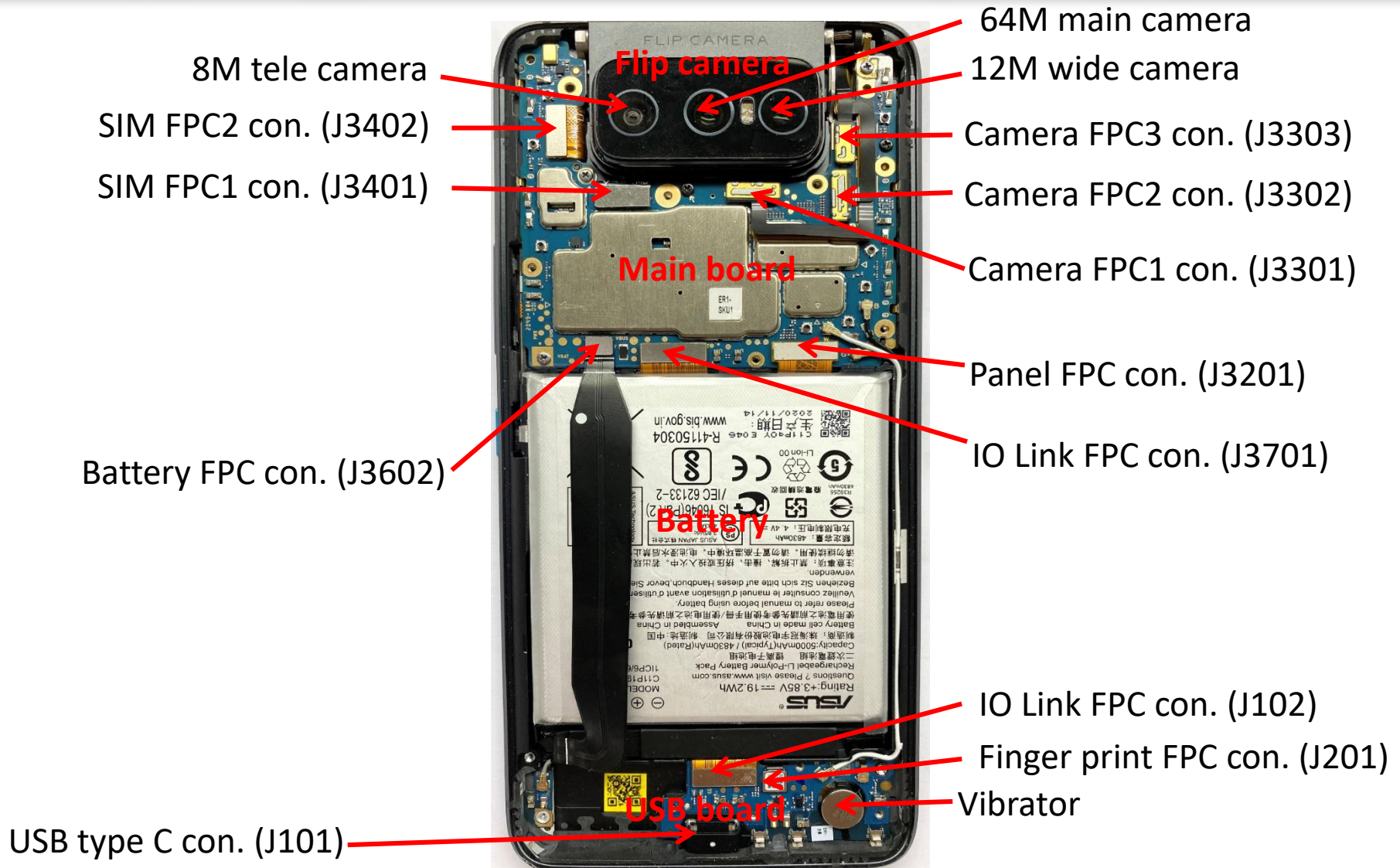
QRD8350



Phone Introduction

Back side (1/2) – MB & USB SB

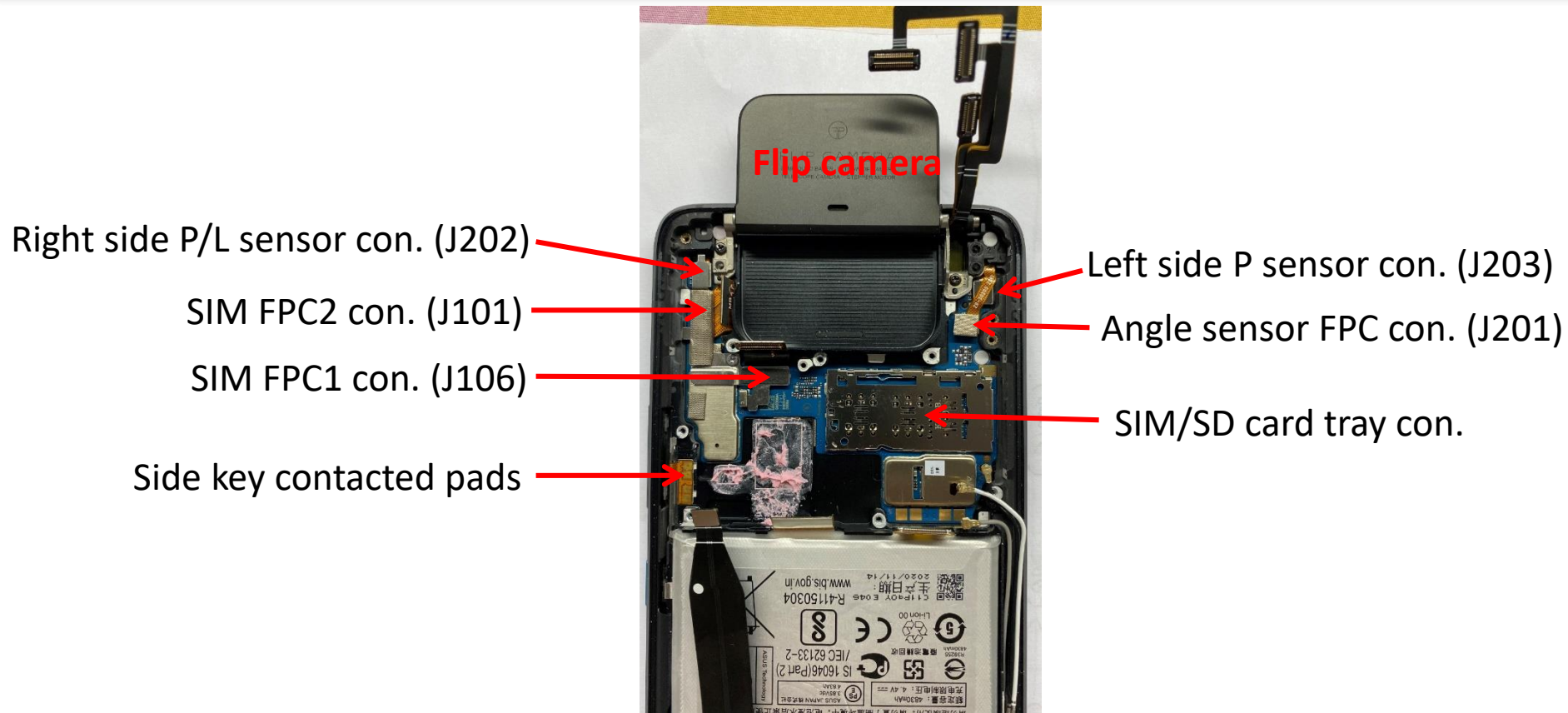
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Phone Introduction

Back side (2/2) – SIM SB

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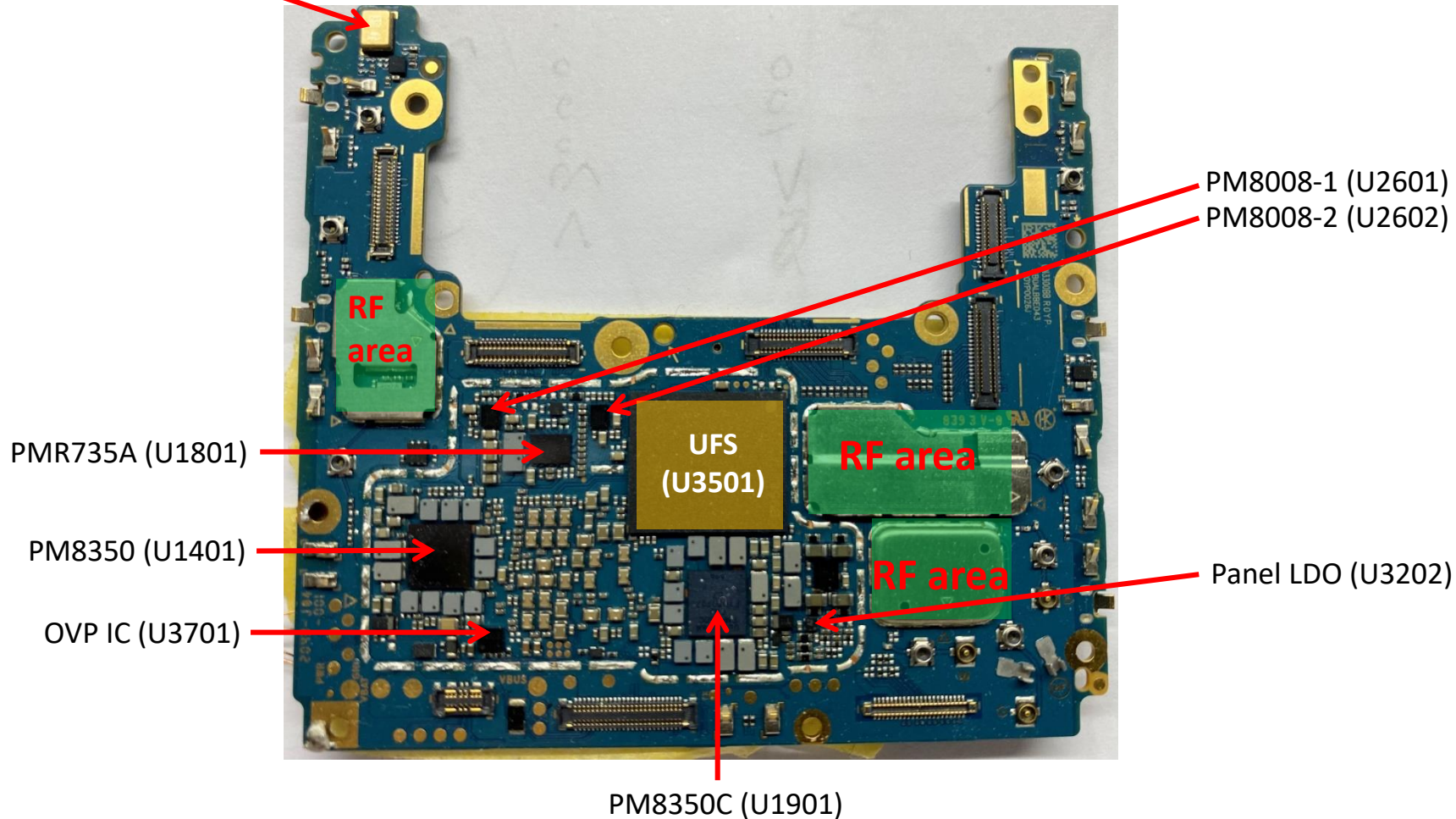


PCB Introduction

MB (1/2) - Bottom

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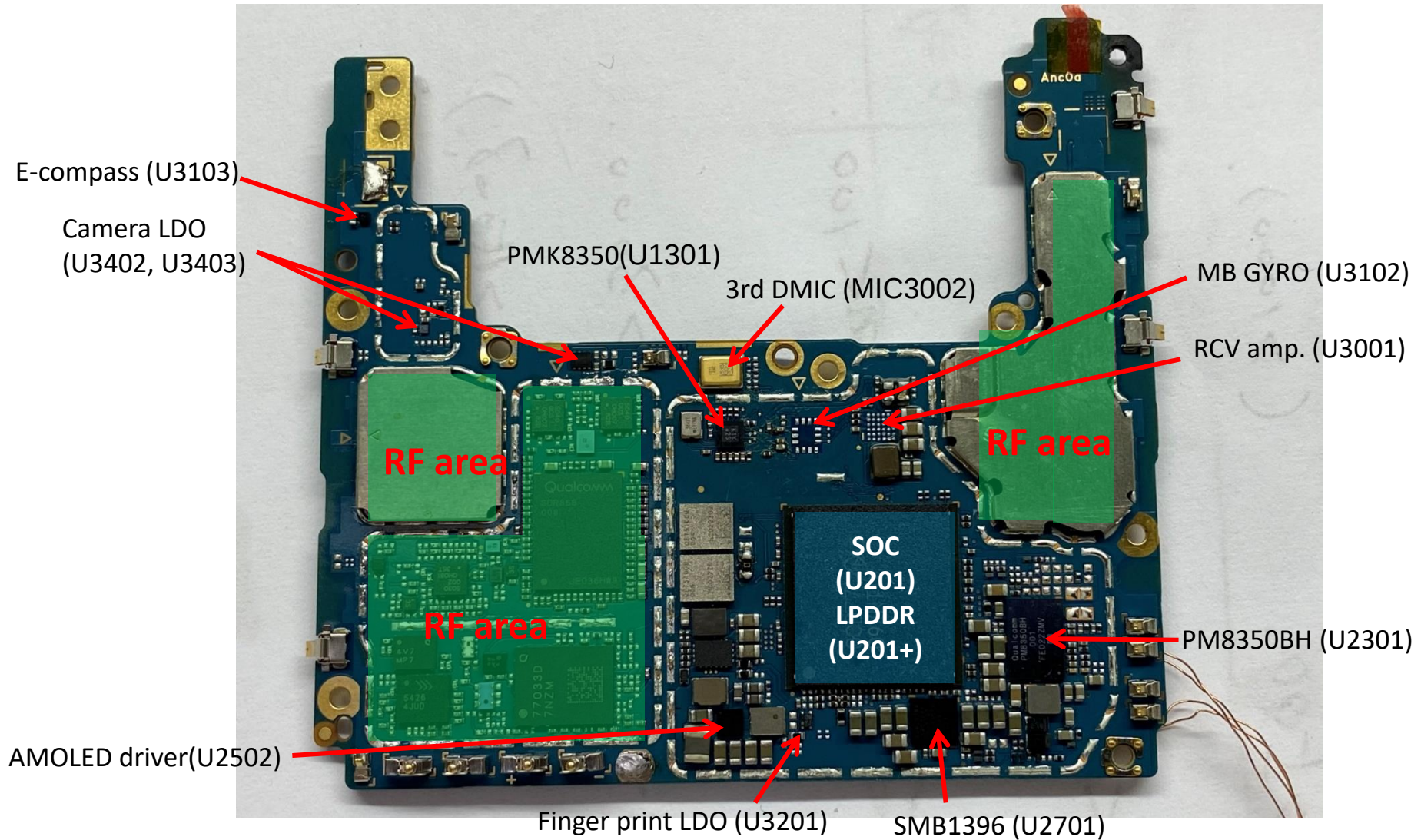
Slave DMIC (MIC3002)



PCB Introduction

MB (2/2) - Top

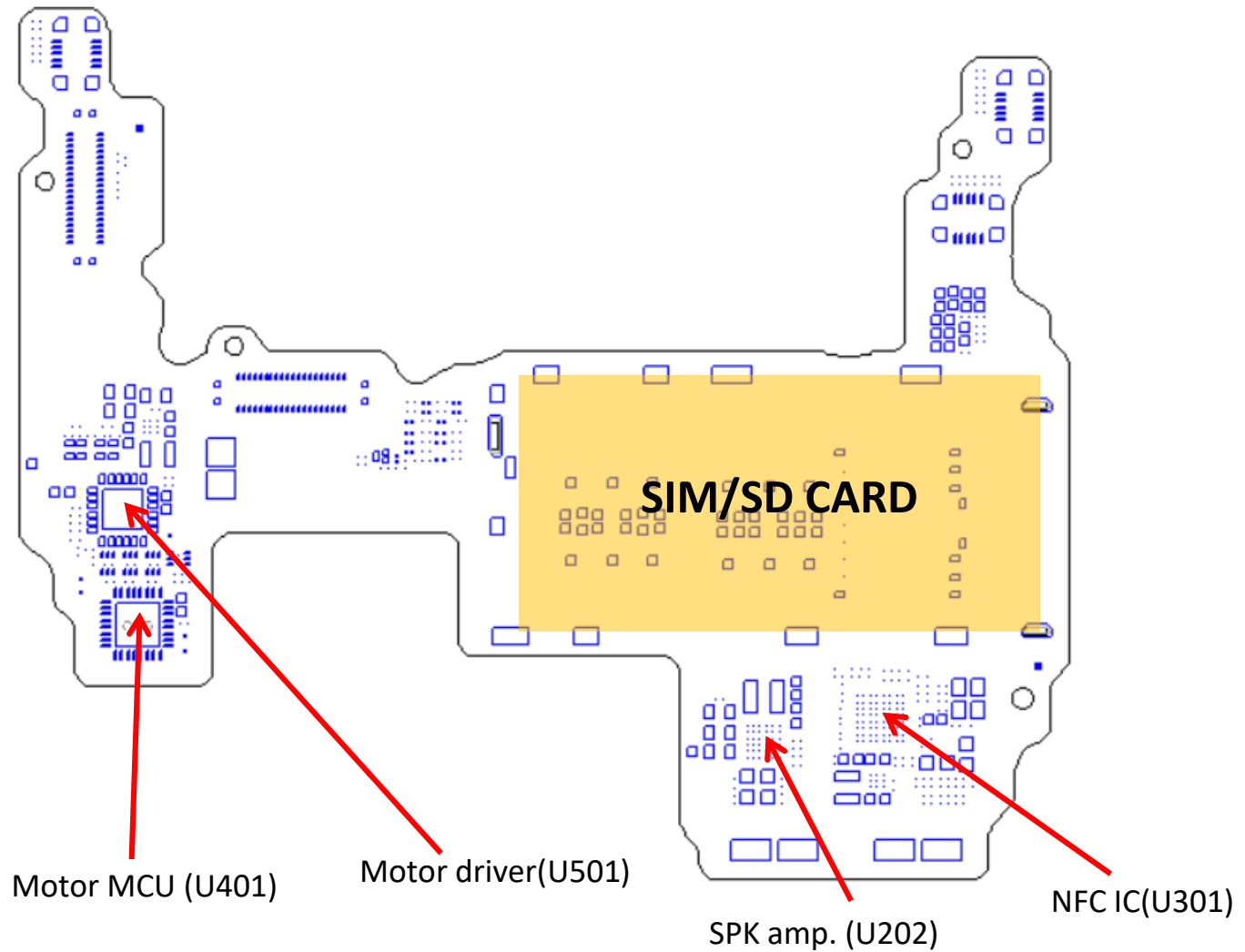
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PCB Introduction

SIM SB Top (1/1)

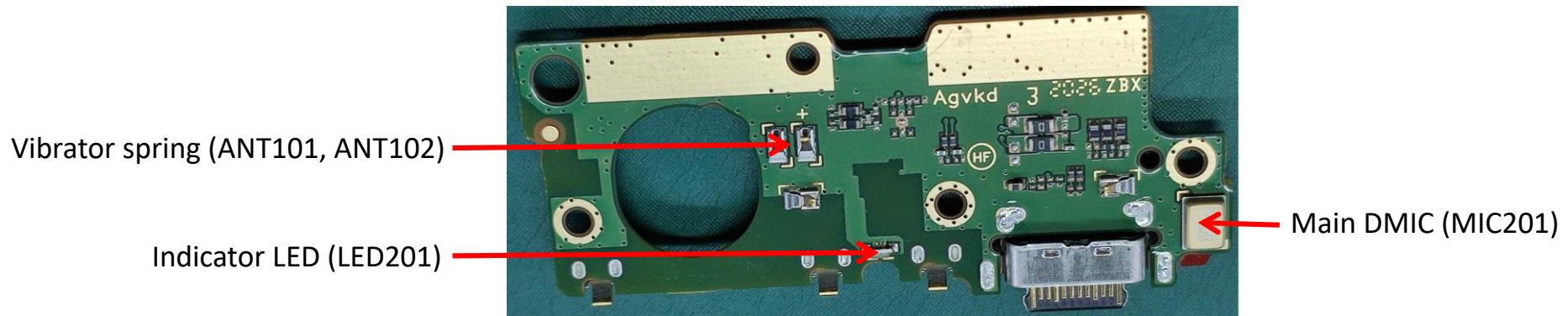
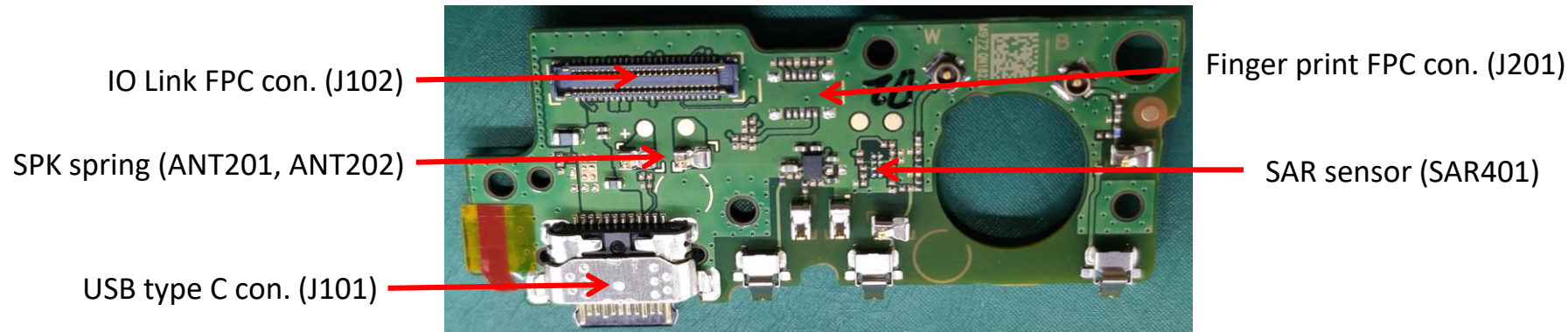
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PCB Introduction

USB SB Top / Bottom (1/1)

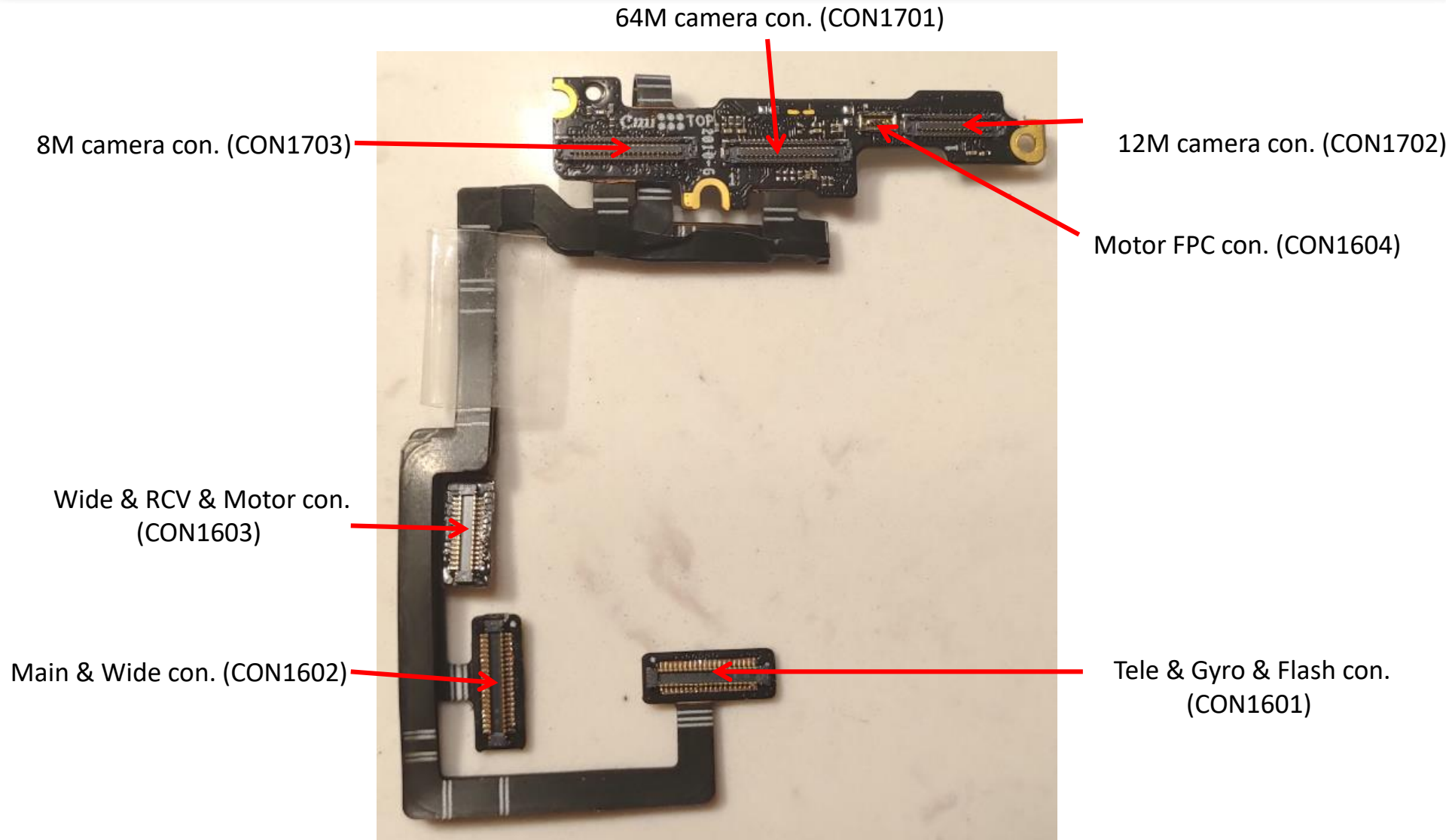
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PCB Introduction

Camera SB Top (1/2)

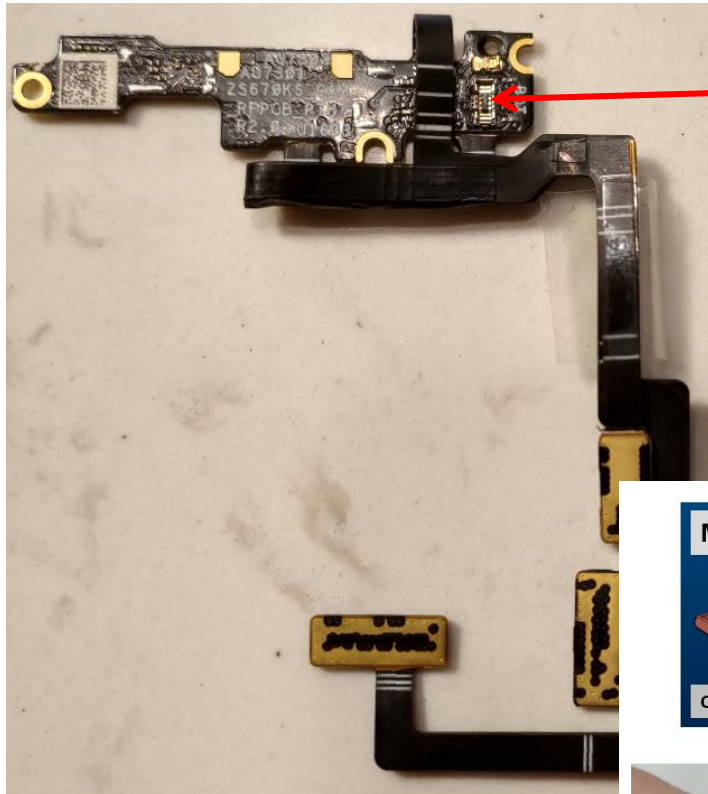
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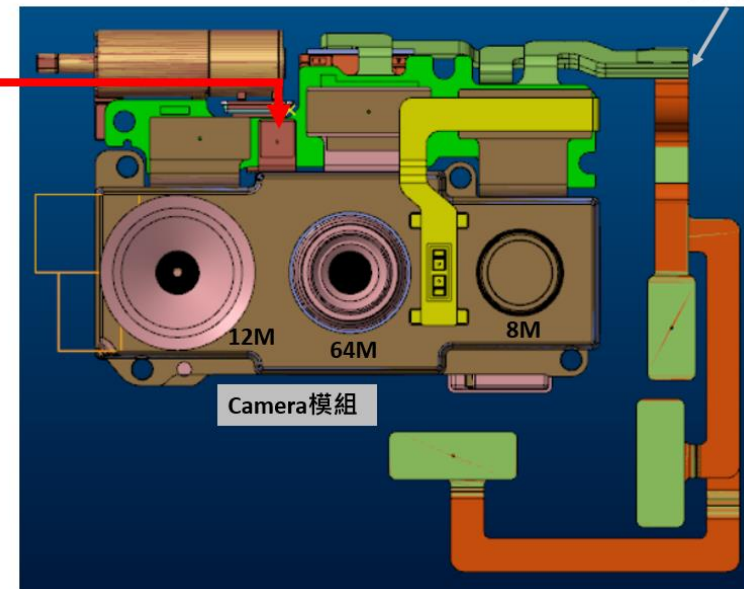
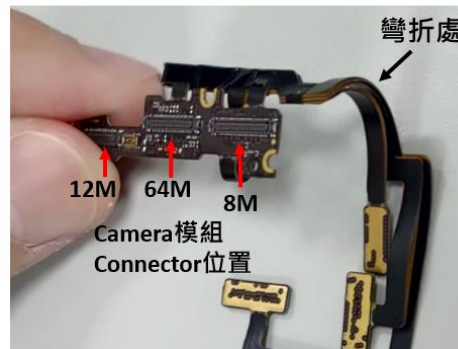
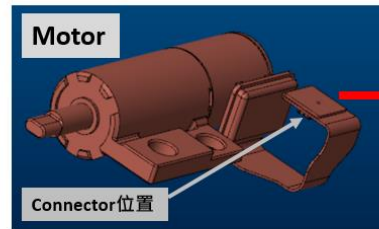
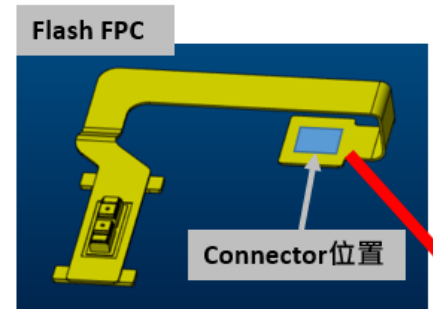
PCB Introduction

Camera SB Bottom (2/2)

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Flash LED con. (CON1605)



- Power on Fail
- UFS Fail
- Panel Fail
- Touch Fail
- 1st G+G Sensor Fail
- 2nd G+G Sensor Fail
- E-compass Sensor Fail
- Proximity Sensor Fail
- Finger Print Fail
- 64M Main Camera Fail
- 12M Wide Camera Fail
- 8M Tele Camera Fail
- SIM/SD Card Fail
- Vibrator Fail
- VOL Key Fail
- Motor Fail
- Indicator LED Fail
- Flash LED Fail
- Receiver Fail
- Speaker Fail
- Main Mic Fail
- Slave Mic Fail
- 3rd Mic Fail

1. Measure the voltage of VBAT (TP3806/ TP3808)

- No voltage:
exchange battery cell;
check bat conn J3602 on MB;
- VBAT's voltage is above 3.4V

if yes, check whether SM_PS_HOLD (TP1304) is 1.8V when the USB is plugged in.

- If PS_HOLD is stabled 1.8V, check whether there is a screen problem or UFS abnormality
- If PS_HOLD can't hold 1.8V, confirm the following:

2. Measure the voltage of VPH_PWR (J3401.43/J3401.44)

- No voltage:
Check whether the following components are connected with tin by X-RAY:
U1401,U1901,U2301,U1801,U2601,U2903,J3401,U5705;
Whether the parallel components and RLC on the VPH_PWR path are abnormal;
U502, J106, U301, U302 on SIM SB
- VPH_PWR's voltage is close to the battery's voltage:
if SM_PS_HOLD is 0V
refer to Power ON Fail (2/3)

➤ Check side KEY

(1) Confirm whether the side KEY FPC is good. (2) While pressing the Power key, confirm whether the voltage of TP3839 is 0V.

If not, check whether the SMT status of ANT3603 or the overlap with key shrapnel is normal

➤ Check Power on sequence

(1) Starting from VREG_BOB, measure the voltage of each power rail in sequence to determine whether the voltage is up. If any power rail is checked to have a short to ground, confirm the SMT status of all components on the path

(2) If one or more groups have no voltage or abnormal voltage, check the SMT status of all components on the power path

(3) reheat or replace the PMIC in order

Reheat : U1401→ U1901→ U1801 →U2301

Replace : U1401→ U1901→ U1801 →U2301

After the hardware boot process is completed, MSM_PS_HOLD(T1911) will be 1.8V. If there is still no display, there are two possibilities:

UFS Fail→Replace CPU/DDR/UFS

OLED Fail-->Check whether the panel con (J3201) and panel are abnormal

Power ON Fail(3/3)

Power on sequence

Event
VREG_COIN
VREG_1P8_SYS
PON trigger is asserted
S1C
BOB
S11B
S12B
L10B
S5B
S10C
S9B
L4B
S2C
S6C_7C
L8B
S10B
GPIO7C
S10C
L6B
VREF_MSM
L5B

Event
L1B
VREG_BB
L1C
L3B
S3C
L2B
L7B
L9B
L6C ¹
L9C ¹
S1B
LNBBCLK1
SLEEP_CLK
PON_RESET_N

1. Check whether the voltage supplied to UFS is normal

Power source	NET name	Check point	Voltage
VCC	VREG_L7B_2P96	C3501.1	2.95
VCCQ1	VREG_L9B_1P2	C3504.1	1.2

2. If the voltage is normal, reheat or replace the IC in sequence

- (i) Reheat U3501 (UFS)
- (ii) Reheat U201 (CPU/DDR)
- (iii) Swap U3501
- (iv) Swap U201

1. Cross-over test,

if the issue follows Panel, check the panel body's cable and BtoB conn ;

if the issue follows MB,

(1).check conn J3201;

(2).check LCD power supply's status;

■ VREG_L13C_3P0_CON: Check J3201.2, C2524, C2525, R2504 to confirm whether it is abnormal.

■ VREG_L12C_1P8_CON : Check J3201.4, R2507, C2531, C2532, C2533 and confirm whether it is abnormal.

■ PANEL_1P2: Check R3269, R3268, C3230 J3201.9 to confirm whether it is abnormal.

■ VREG_DIS_-2P7_CON: Check J3201.3, R2506, C2529, C2530 to confirm whether it is abnormal

■ VREG_DIS_4P6_CON: Check J3201.43/41/39, R2503, C2523, C2522 to confirm whether it is abnormal

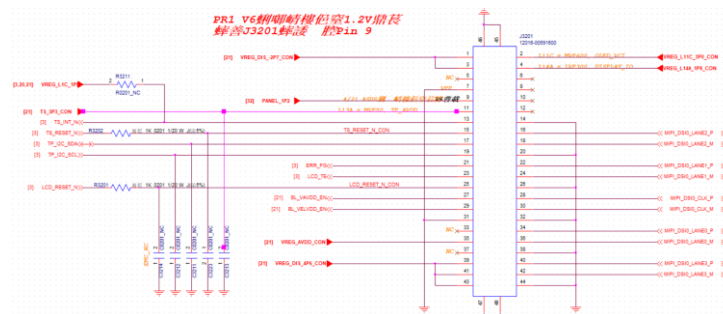
■ VREG_AVDD_CON: Check J3201.35, R2508, C2539 , C2540 to confirm whether it is abnormal

If the above components are confirmed to be no abnormality, but one group\multiple groups of voltages are still abnormal, please Reheat or replace U2502

Touch Fail

Firstly cross-over test to confirm the panel is a good product

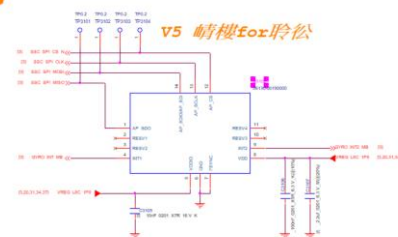
1. Check the SMT status of J3201.
2. Measure the net TP_I2C_SDA & TP_I2C_SCL's voltage status, they should be 1.8V, if not, check R401 & R402 SMT status.
3. Confirm TS_RESET_N status, check R3202, U201 X-ray to confirm whether it is abnormal.
4. Confirm the status of TS_INT_N. If there is an abnormality, it is a U201 problem.
5. Confirm TS_3P3_CON status, check R2505, C2526, C2527, U1901, J3201.11;
6. If no problems are found in the above, please reheat U1401&U201 in order first, if it is invalid, replace U1401&U201 in order.



1st G+G Sensor Fail

1. Confirm whether VREG_L8C_1P8 has power supply, and measure C3105.2.
2. If there is a short circuit to ground or no voltage, first confirm the state of the capacitor and resistor connected to VREG_L8C_1P8.
3. If none of the above components are abnormal, but VREG_L8C_1P8 has no voltage, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, reheat U1901, if it is invalid, replace U1901
4. If VREG_L8C_1P8 has voltage, take x-ray to U3102 to confirm whether it is correctly mounted.
5. Measure whether the SSC_SPI_CLK, SSC_SPI_MOSI, SSC_SPI_MISO, SSC_SPI_CS_N and GYRO_INT1/2 signals are normal.
6. After finishing the above tests, it can't work, just replace U3102.
7. The above 1~6 points are the MB G-sensor confirmation process, and the G-sensor on the camera SB shall be handled by comparison

A+G



1. First confirm whether MB J3301, J3302, and J3303 all correspond to camera SB CON1601, CON1602, CON1603 buckle normally, and determine whether the problem occurs in CAM SB or MB
2. ■ If the problem occurs in CAM SB
3. Check the SMT status of CAM SB CON1703.
4. Measure whether R1724 is 2.9V, if not, check whether C1718.1 is shorted to ground.
5. Measure whether R1722 is 1.8V, if not, check whether C1716.1, C1712.1, C1707.1 are shorted to ground.
6. If none of the above is abnormal, please replace the 8M tele module, if it is invalid, replace the CAM SB.
7. ■ If the problem occurs in MB
8. Check the SMT status of MB J3301, J3302, J3303, C3338, C3301, C3330, R3301, C3318, C2005.
9. If none of the above is abnormal, first reheat MB U1901, U201, if it is invalid, replace U1901, U201.

E-compass Sensor Fail

1. Confirm whether VREG_L8C_1P8 has power supply, and measure C3103.1.
2. If there is a short circuit to ground or no voltage, first confirm the state of the capacitor connected to VREG_L8C_1P8.
3. If none of the above components are abnormal, but VREG_L8C_1P8 has no voltage, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, reheat U1901, if it is invalid, replace U1901.
4. If VREG_L8C_1P8 has voltage, take x-ray to U3103 to confirm whether it is correctly mounted.
5. Measure whether the SSC_I2C1_SCL, SSC_I2C1_SDA signals are normal.
6. After completing the above tests, you can't work, so replace U3103 directly.

Proximity Sensor Fail

1. Confirm whether SIM FPC1 J106, SIM FPC2 J101 are properly buckled with the mainboard, and check the SMT status.
2. Make sure that the two P sensors of SIM SB are connected to J202 and J203 normally.
3. Confirm which P sensor is abnormal J202 or J203

■ SIM SB J202:

1. View the SMT status of J202, R211, C212, C214, C216, C218, C228.
2. If there is no abnormality, measure whether there is 3.0V voltage at both ends of R211. If there is no abnormality, check whether the mainboard C2003.1 has 3.0V voltage. If there is no abnormality, reheat U1901 and replace U1901 invalid.
3. If the voltage is not abnormal, then confirm whether the AL/P_I2C_SDA_1/AL/P_I2C_SCL_1 path signal is abnormal.
4. If there is no abnormality, replace the P sensor connected to J202. If it is invalid, reheat U201 and replace U201 again.

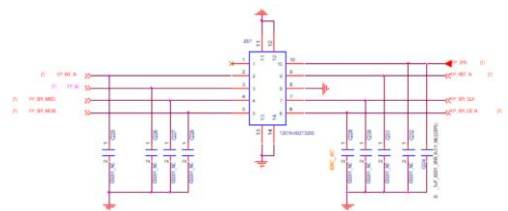
■ SIM SB J203:

1. Check the SMT status of J203, R212, C213, C215, C217, C219, C226.
2. If there is no abnormality, measure whether there is 3.0V voltage at both ends of R212. If there is no abnormality, check if there is 3.0V voltage on the mainboard C2003.1. If there is no abnormality, reheat U1901 and replace U1901 invalid.
3. If the voltage is not abnormal, then confirm whether the AL/P_I2C_SDA_2/ AL/P_I2C_SCL_2 path signal is abnormal.
4. If there is no abnormality, replace the P sensor connected to J202. If it is invalid, reheat U201 and replace U201 again.

Finger Print Fail

1. Confirm that the connection between USB BD. J201 and FP unit is normal.
2. Confirm whether SUB BD. C232.1 has 3.0V voltage.
3. If there is a short circuit to ground or no voltage, first confirm the SMT status of U3201, C3229, C3228.
4. If none of the above components are abnormal and FP_3P0 has no voltage, confirm whether VREG_BOB has electricity, if it is short-circuited, find the short-circuit point and repair it, if it is not short-circuited, first reheat U1901, or if it is invalid, change U1901
5. If FP_3P0 has voltage, check J3202 whether the pin is abnormal.
6. Measure whether FPS_SPI_MOSI & FPS_SPI_MISO & FPS_SPI_CLK & FPS_SPI_CS_N & FPS_RST_N & FPS_INT_N & FP_ID signals are normal.
7. After the above tests are completed, the work cannot be done. Priority is given to replacing the FP with good products, and then reheating U201 if it is invalid, or replacing U201 if it is still invalid.

FP



Angel sensor fail

1. Check whether the SIM board connector J201 has snapping problems or damage.
2. Check whether the Angel sensor FPC is broken or abnormal, and whether the IC is damaged.
3. Swap Angel sensor FPC/SIM board/SIM_Left FPC/MB confirms the phenomenon follower.

If it is connected to the SIM board or mainboard :

1. Confirm that R208.2 has a 1.8V voltage. If not, confirm the SIM board J101.36,J106.22 status
2. Measure whether the signals ANGLE_INT_N_1P8, ANGLE_I2C_SDA_1P8, ANGLE_I2C_SCL_1P8 are abnormal, and troubleshoot the components on the abnormal signals to replace the problem components.

64M Main Camera Fail

1. Confirm whether the connector (J3302) of Camera RFPC is fully buckled on the MB.
2. Check the connector (J3302) for damage or foreign objects.
3. Swap all keypart confirmation phenomena are related to MB or RFPC.

If the Swap phenomenon is related to MB :

1. Measure whether the impedance of CAM0_DVDD_1P1_MB/ CAM0_AVDD_2P9_MB/ CAM0_AVDD_1P8_MB/CAM02_DRV_AVDD_2P9_MB is abnormal, if it is abnormal, check the component U2210 on the circuit, and replace U2210 if there is a problem.
2. Measure whether the impedance of CAM0_VM_3P1_MB is abnormal, if it is abnormal, check the components C3403, U3402 on the line, if there is a problem, reheat or replace the component (C3403, U3402)
3. Measure CAM012_GYRO_EEPROM_DOVDD_1P8_MB if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

If the swap phenomenon is similar to RFPC

1. Check the connector (CON1602) for damage or foreign objects.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and the 64M camera module or RFPC

→ If the Swap phenomenon is related to RFPC :

1. Measure whether the impedance of CAM0_DVDD_1P1_MB/ CAM0_AVDD_2P9_MB/ CAM0_AVDD_1P8_MB/ is abnormal, if abnormal, check the components on the circuit and replace them
2. Check whether the D-phy signal has an open circuit or an abnormal impedance (CAM_MCLK0/CSI0_A0, B0, C0/CSI0_A1, B1C1/CSI0_A2, B2, C2)

12M Wide Camera Fail

1. Check whether the connector (J3302&J3303) of Camera RFPC is fully buckled on the MB.
2. Check whether the connector (J3302&J3303) is damaged or foreign.
3. Swap all keypart confirmation phenomena are related to MB or RFPC.

If Swap phenomenon is related to MB :

1. Measure whether the impedance of CAM1_VDDD_1P05_MB/ CAM1_VDDA_2P8_MB is abnormal, if it is abnormal, check the component U2210 on the line, if there is a problem, replace U2210.
2. Measure CAM1_VDD_AF_2P8_MB/CAM012_GYRO_EEPROM_DOVDD_1P8_MB, if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

If the swap phenomenon is similar to RFPC:

1. Check the connectors (CON1602, CON1603) for damage or foreign objects.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and 12M camera module or RFPC

→ If Swap phenomenon is related to RFPC :

1. Measure whether the impedance of CAM1_VDDD_1P05_MB/ CAM1_VDDA_2P8_MB / CAM1_VDD_AF_2P8/CAM012_GYRO_EEPROM_DOVDD_1P8_MB is abnormal, if it is abnormal, check the components on the circuit and replace them
2. Check whether the MIPI signal impedance is abnormal or abnormal (CAM_MCLK1/CSI1_LANE0_P, M/CSI1_LANE1_P, M/CSI1_LANE2_P, M/CSI1_LANE3_P, M/CSI1_CLK_M, P)

8M Tele Camera Fail

1. Confirm whether the connector (J3301) of Camera RFPC is fully buckled on the MB.
2. Check whether the connector (J3301) is damaged or foreign.
3. Swap all key part confirmation phenomena are related to MB or RFPC.

If Swap phenomenon is related to MB :

1. Measure whether the impedance of CAM2_VM_2P8_MB is abnormal. If it is abnormal, check the component U3403 on the line. If there is a problem, replace U3403.
2. Measure CAM2_AVDD_2P8_MB/CAM02_DRV_AVDD_2P9_MB, if it is abnormal, check the component U2210 on the line, if there is a problem, replace U2210.
3. Measure whether the impedance of CAM2_DVDD_1P2_MB_CON is abnormal, if it is abnormal, check the component U2301 on the line, if there is a problem, replace U2301.
4. Measure CAM012_GYRO_EEPROM_DOVDD_1P8_MB if it is abnormal, check the component U1801 on the line, if there is a problem, replace U1801.

If the swap phenomenon is similar to RFPC:

1. Check the connector (CON1601) for damage or foreign matter.
2. Check whether the RFPC itself is damaged (Ex. scratches, screw locks)
3. Turn on the camera module, confirm the phenomenon and 8M camera module or RFPC

→ If Swap phenomenon is related to RFPC :

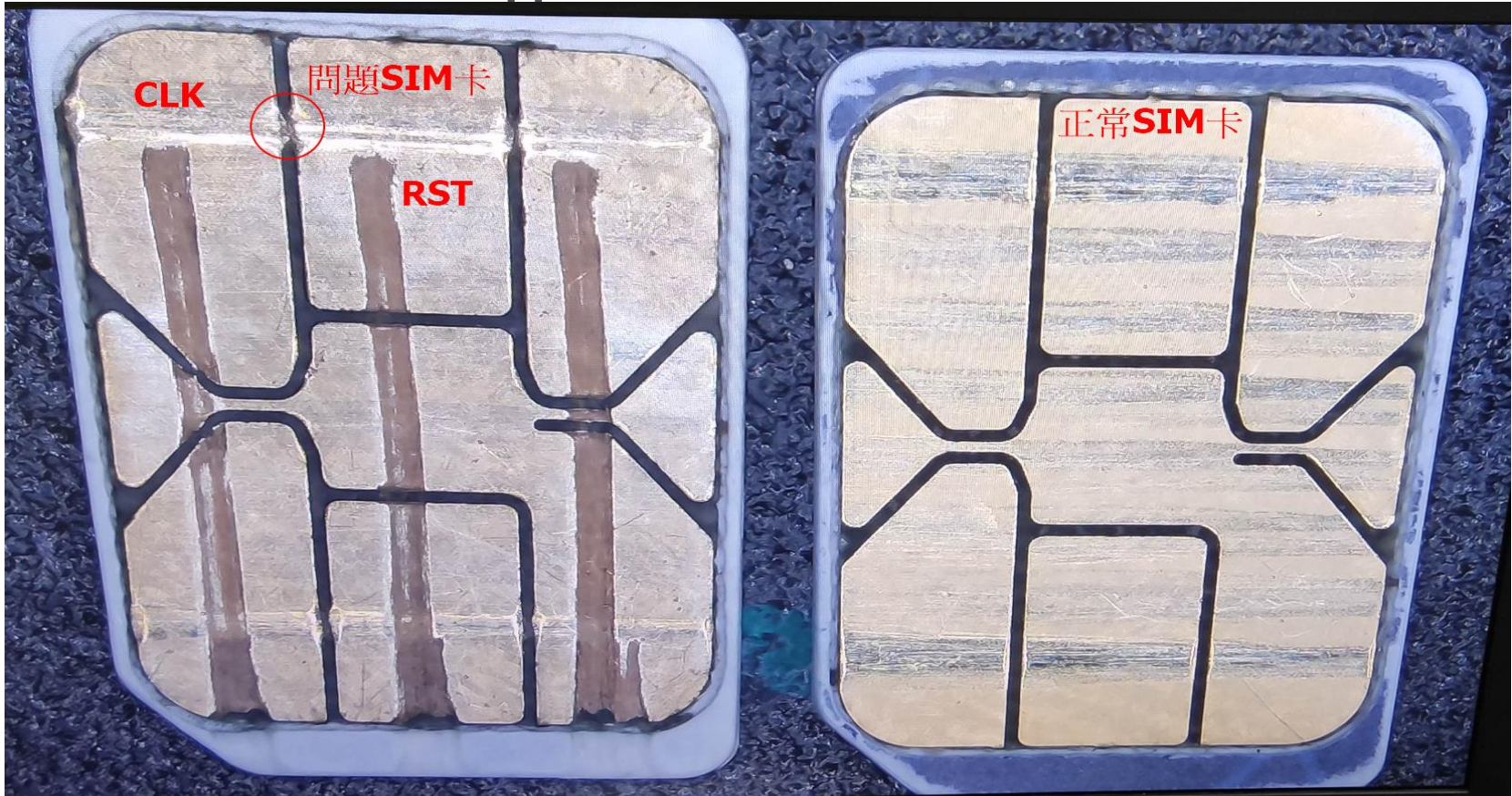
1. Measure whether the impedance of CAM2_VM_2P8 / CAM2_DVDD_1P2 / CAM2_DOVDD_1P8/CAM2_AVDD_2P8/CAM2_DRV_AVDD_2P9 is abnormal, if it is abnormal, check the components on the circuit and replace them
2. Check whether the MIPI signal impedance is abnormal or if the impedance is abnormal (CAM_MCLK2/ CSI2_LANE0_M, P/ CSI2_LANE1_M, P/ CSI2_LANE2_M, P/ CSI2_LANE3_M, P/ CSI2_CLK_M, P)

SIM/SD Card Fail (1/3)

1. Make sure that SIMFPC_MID, SIM tray, SIM 1/2, SD card are normal, check the SMT status of SIM SB J01~J05, J106, MB J3401.
2. SIM 1 fail, measure whether SIM SB R108 has power supply (VCC_NFC1), whether R107 has power supply (VREG_L4C_1P8), and whether R110 (UIM1_RESET) is abnormal.
3. SIM 2 fail, measure whether SIM SB R114 has power supply (VCC_NFC2), whether R115 has power supply (VREG_L5C_1P8), and whether R118 (UIM2_RESET) is abnormal.
4. SD card fail, measure whether SIM SB C2514.1 has power supply (VREG_L9C_2P96). If not, check SIM SB J106, D101.
5. Check whether the SIM SB R123 will be pulled from 0V to 1.8V after the SIM tray is inserted, and judge whether the DET pin is active. If not, check the SMT status of SIM SB R123, R122, Q102, R124, R125, J105.
6. Check SDC2_CLK_CONN with the SD card inserted, and measure MB to see if there is a CLOCK. If not
7. Check UIM1_CLK/UIM2_CLK with dual SIM inserted, and measure whether SIM R113 and R120 have CLOCK. If not, check the SMT status of MB
8. If there is no problem above, check whether the SIM card used is bad

SIM/SD Card Fail (2/3)

Observe whether there is a short circuit or abnormality in the appearance of the SIM card



SIM/SD Card Fail (3/3)

Measure the impedance between each pin

The following is a reference value (China Mobile SIM card),
please refer to the value of most SIM cards on the production line

	Red prob(+)						
		C1	C2	C3	C5	C6	C7
Block prob(-)	C1		6.4M	6.4M	1.5M	open	60.7k
	C2	9.3M		33M	1.9M	open	9.3M
	C3	13.2M	38.7M		2M	open	13.3M
	C5	7.2M	21.8M	22.1M		open	7.3M
	C6	open	open	open	open		open
	C7	60.6K	6.5M	6.5M	1.6M	open	

【 SIM Card Pin Assignments 】

PIN	Name
C1	Vcc
C2	RST
C3	CLK
C4	Reserved
C5	GND
C6	Vpp
C7	I/O
C8	Reserved



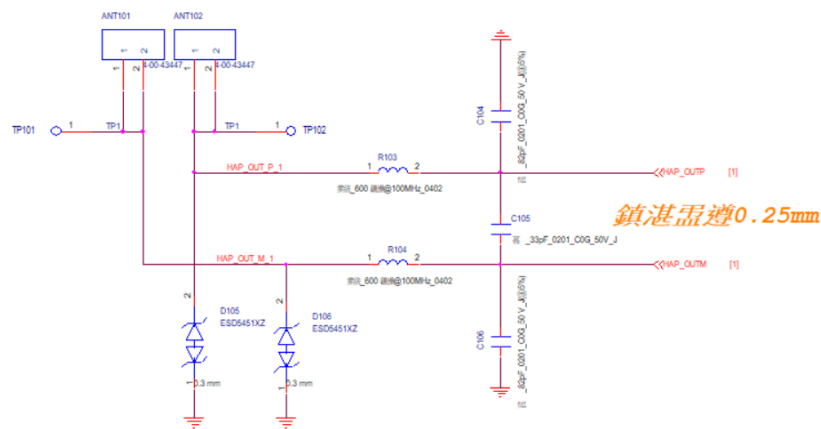
Vibrator Fail

1. Check whether the IO FPC/USB small card connector (J3701, J102) has no buckles or foreign objects or shrapnel is abnormal, and whether there is pressure on the assembly.
2. Measure whether the normal PM impedance of the Vibrator unit is normal without short circuit.
3. Swap mainboard/LKFPC/USB small card to find the problem follow point

If with LKFPC → Replace LKFPC

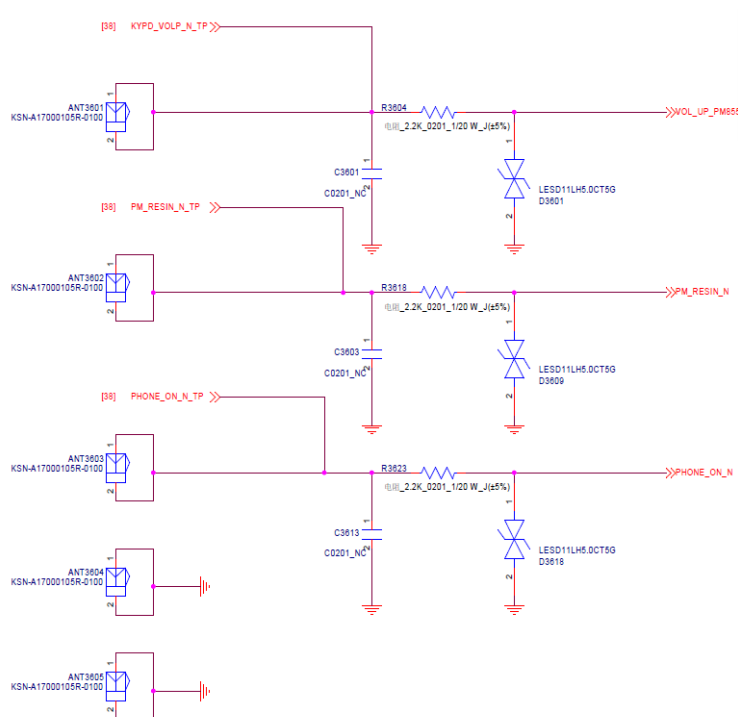
If with USB board → Check R103, R104, ANT101, ANT102 and other RLC components SMT status ANT101

If it comes with the mainboard → Check U2301(PM8350BH) SMT status, first reheat U2301, if it is invalid, replace U2301.



VOL Key Fail

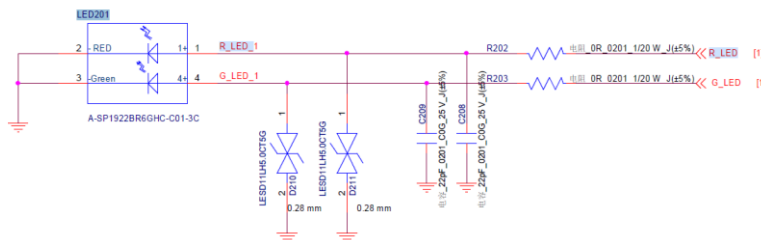
1. Check if ANT3601, ANT3602, ANT3603, ANT3605 and Side key FPC are connected properly.
2. Check the SMT status of MB R3604, R3618, R3623, D3601, D3609, D3618.
3. Press the "DOWN" button to check whether MB C3603.1 has pulled LOW from HIGH.
4. Press the "UP" button to check whether MB C3601.1 pulls LOW from HIGH.



Indicator LED Fail

1. Check the contact status of MB J3701 with USB SB J102, , then confirm problems occurred in USB SB or MB.
- If problems occurred in USB SB:
 1. Check the SMT status of J102, R202, R203, C208, C209, D210, D211 and LED201.
 2. If the above steps are confirmed normal, but the failure is still existing, then swap the LED201.
- If problems occurred in MB:
 1. Check the SMT status of MB J3701 and U1901.
 2. If the above steps are confirmed normal, but the failure is still existing, reheat MB U1901 first, then swap the MB U1901.

INDICATOR



Flash LED Fail

1. Check the contact status of MB J3301 with camera FPC CON1601, then confirm problems occurred in Flash FPC, CAM SB or MB.
 - If problems occurred in CAM SB:
 1. Check the SMT status of Flash FPC C1601, C1602 and CON1605.
 2. Check the voltage of C1601.1 is 3V, if not, check the SMT status of C1601 and C1602.
 3. If the above steps are confirmed normal, but the failure is still existing, then swap the Flash FPC.
 - If problems occurred in MB:
 1. Check the SMT status of MB J3301, U1901
 2. Check the voltage of FLASH_LED1, FLASH_LED2
 3. If the above steps are confirmed normal, but the failure is still existing, reheat MB U1901 first, then swap the U1901.

Receiver Fail

1. Check the SMT and contact status of Receiver, camera FPC CON1603 and MB J3303.
2. Check the SMT and contact status of CAM SB AU7301 and AU7302 between Receiver.
3. Check the SMT status of MB R3003, R3004, R3005, R3006, L3002, L3003, U3001.
4. Check the voltage of MB C3012.2 is 1.8V. If is not, check the SMT status.
5. Check the voltage of MB C3001.1 is normal. If not, check the SMT status of L3001.
6. If the above steps are confirmed normal, but the failure is still existing. Reheat MB U3001 first, then swap the MB U3001.

Speaker Fail

1. Check the SMT and contact status of Speaker, IO FPC, SIM FPC1 and SIM FPC2.
2. Check the USB SB ANT201 and ANT202 contact status with Speaker.
3. Check the SMT status of USB SB: B201, B202, J102 , SIM SB: R202, R203, R204, R205, B201, B202, U202. MB: J3701, ANT3001, ANT3002.
4. Check the voltage of SIM SB C211.2 is 1.8V. If is not, check the SMT and contact status of FPCs between SIM board and MB.
5. Check the voltage of SIM SB C201.1 is normal. If is not, check the SMT status of L201. If the above steps are confirmed normal, but the failure is still existing. Reheat SIM SB U202 first, then swap the SIM SB U202.

Main Mic Fail

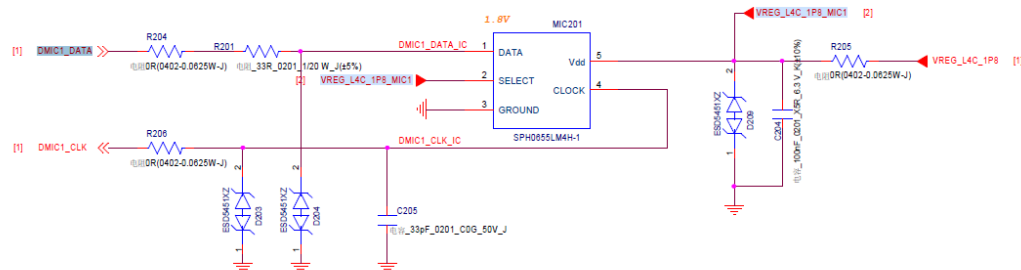
■ If problems occurred in MB:

1. Check the SMT status of MB J3701, R326.
2. Check the voltage of C2008.1 is 1.8V, if is not or abnormal, reheat U1801 first, then swap the U1801.

■ If problems occurred in USB SB:

1. Check the SMT status of USB SB J102
2. Check the SMT status of USB SB MIC201, R204, R205, R206, R201, D203, D204, C205, C204, D209.
3. Check the voltages of USB SB R205.1 / R205.2 are 1.8V.
4. If the above steps are confirmed normal, but the failure is still existing.
5. Reheat the USB SB MIC201 first, then swap the MB MIC201.

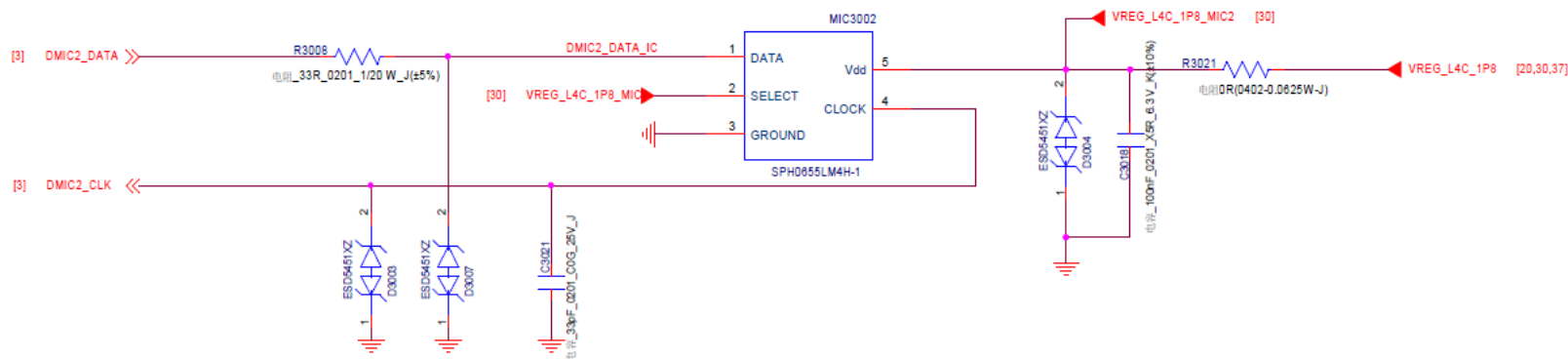
Main DMIC



Slave Mic Fail

1. Check the voltage of MB R3018.1 is 1.8V, if is not or abnormal, reheat U1901, the failure is still existing, please swap the U1901.
 2. Check the SMT status of MB MIC3002, R327, R3008, D3003, D3004, D3007, C3018, C3021, R3021.
 3. Check the voltages of MB R3021.1 / R3021.2 are 1.8V.
- If the above steps are confirmed normal, but the failure is still existing.
reheat MB MIC3002 first, then swap the MB MIC3002.

Slave DMIC



1. Check the voltage of MB C3029.1 whether there is power supply, if is not or abnormal, reheat U1901, the failure is still existing, please swap the U1901.
2. Check the SMT status of MB MIC3001, R3014, D3005, D3006, D3008, C3029
3. Check the voltages of MB R3017.1/ R3017.2 are 1.8V ◦
4. If the above steps are confirmed normal, but the failure is still existing.
reheat MB MIC3001 first, and then swap MB MIC3001 ◦

Thank you!